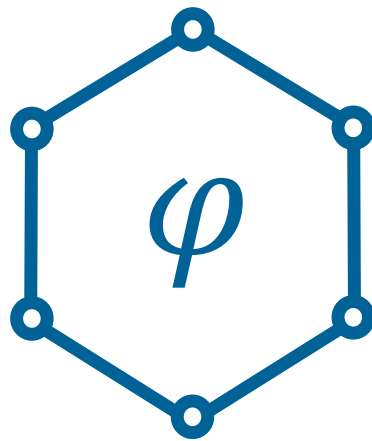


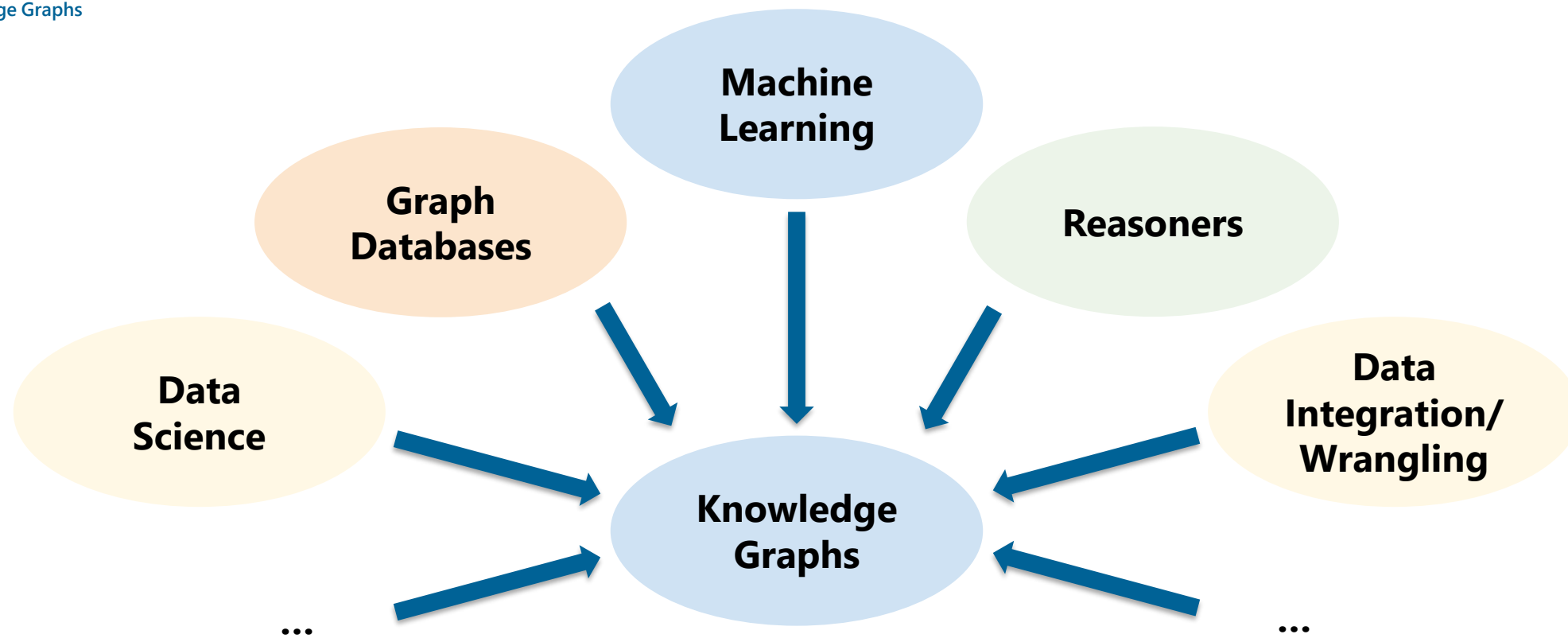


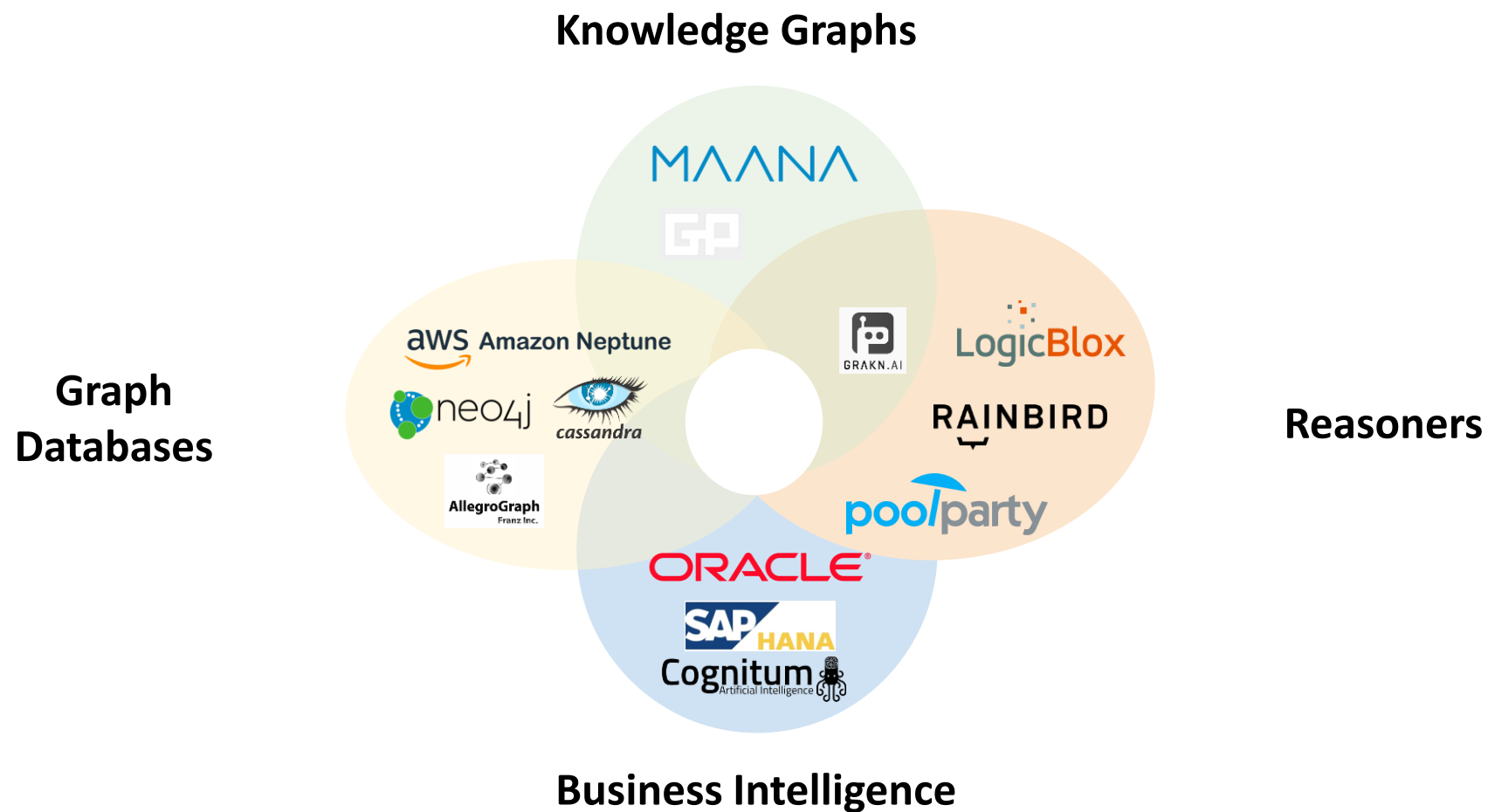
Logical Knowledge in KGs

Theory to Practice

Emanuel Sallinger









Semantic Web

(Relational) Databases

Reasoners

Graph Databases



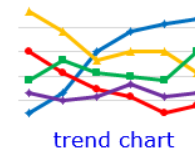


Databases

DB-Engines Ranking

The DB-Engines Ranking ranks database management systems according to their popularity. The ranking is updated monthly.

Read more about the [method](#) of calculating the scores.



370 systems in ranking, April 2021

Rank	Rank			DBMS	Database Model	Score		
	Apr 2021	Mar 2021	Apr 2020			Apr 2021	Mar 2021	Apr 2020
1.	1.	1.	1.	Oracle +	Relational, Multi-model ⓘ	1274.92	-46.82	-70.51
2.	2.	2.	2.	MySQL +	Relational, Multi-model ⓘ	1220.69	-34.14	-47.66
3.	3.	3.	3.	Microsoft SQL Server +	Relational, Multi-model ⓘ	1007.97	-7.33	-75.46
4.	4.	4.	4.	PostgreSQL +	Relational, Multi-model ⓘ	553.52	+4.23	+43.66
5.	5.	5.	5.	MongoDB +	Document, Multi-model ⓘ	469.97	+7.58	+31.54
6.	6.	6.	6.	IBM Db2 +	Relational, Multi-model ⓘ	157.78	+1.77	-7.85
7.	7.	8.	8.	Redis +	Key-value, Multi-model ⓘ	155.89	+1.74	+11.08
8.	8.	7.	7.	Elasticsearch +	Search engine, Multi-model ⓘ	152.18	-0.16	+3.27
9.	9.	9.	9.	SQLite +	Relational	125.06	+2.42	+2.87
10.	10.	10.	10.	Microsoft Access	Relational	116.72	-1.41	-5.19
11.	11.	11.	11.	Cassandra +	Wide column	114.85	+1.22	-5.22
12.	12.	12.	12.	MariaDB +	Relational, Multi-model ⓘ	96.37	+1.92	+6.47
13.	13.	13.	13.	Splunk	Search engine	88.49	+1.56	+0.41
14.	14.	14.	14.	Hive	Relational	78.50	+2.46	-5.56
15.	16.	23.	23.	Microsoft Azure SQL Database	Relational, Multi-model ⓘ	71.84	+0.96	+32.89
16.	17.	16.	16.	Amazon DynamoDB +	Multi-model ⓘ	70.73	+1.84	+6.46
17.	15.	15.	15.	Teradata	Relational, Multi-model ⓘ	70.55	-0.88	-6.04
18.	20.	18.	18.	SAP HANA +	Relational, Multi-model ⓘ	53.44	+2.45	+0.15
19.	19.	19.	19.	SAP Adaptive Server	Relational, Multi-model ⓘ	51.67	-0.51	-0.96
20.	18.	21.	21.	Neo4j +	Graph, Multi-model ⓘ	51.04	-1.28	+0.24





Graph Databases

☐ include secondary database models

32 systems in ranking, April 2021

Rank			DBMS	Database Model	Score		
Apr 2021	Mar 2021	Apr 2020			Apr 2021	Mar 2021	Apr 2020
1.	1.	1.	Neo4j	Graph, Multi-model	51.04	-1.28	+0.24
2.	2.	2.	Microsoft Azure Cosmos DB	Multi-model	33.52	+1.11	+1.47
3.	3.	3.	ArangoDB	Multi-model	4.77	-0.28	-0.11
4.	4.	4.	OrientDB	Multi-model	4.23	-0.48	-0.29
5.	5.	5.	Virtuoso	Multi-model	3.18	+0.30	+0.56
6.	7.	8.	GraphDB	Multi-model	2.24	-0.03	+1.08
7.	6.	7.	JanusGraph	Graph	2.10	-0.33	+0.32
8.	8.	6.	Amazon Neptune	Multi-model	1.90	+0.04	+0.09
9.	10.	10.	Stardog	Multi-model	1.63	+0.13	+0.67
10.	11.	12.	TigerGraph	Graph	1.57	+0.07	+0.69
11.	9.	13.	Fauna	Multi-model	1.52	-0.31	+0.65
12.	12.	14.	AllegroGraph	Multi-model	1.35	+0.04	+0.52
13.	13.	9.	Dgraph	Graph	1.16	-0.08	+0.04
14.	14.	11.	Giraph	Graph	1.13	+0.01	+0.19
15.	15.	23.	Nebula Graph	Graph	0.99	+0.02	+0.83
16.	16.	15.	Blazegraph	Multi-model	0.80	-0.03	+0.16
17.	17.	17.	Grakn	Multi-model	0.76	+0.04	+0.35
18.	18.	16.	Graph Engine	Multi-model	0.70	-0.01	+0.14
19.	19.	18.	InfiniteGraph	Graph	0.51	+0.01	+0.11
20.	20.	19.	FlockDB	Graph	0.33	+0.01	+0.06





Datalog Reasoners

Written in	Name	Try it online	External Database	Description	Licence
C	XSB			A logic programming and deductive database system for Unix and MS Windows with tabling giving Datalog-like termination and efficiency, including incremental evaluation ^[16]	GNU LGPL
	Coral ^[17]			A deductive database system written in C++ with semi-naïve datalog evaluation. Developed 1988-1997.	custom licence, free for non-commercial use
C++	DLV ^[18]			A Datalog extension that supports disjunctive head clauses.	custom licence, free for academic and non-commercial educational use, as well as for use by non-profit organisations ^[18]
	Inter4QL ^[20]			an open-source command-line interpreter of Datalog-like 4QL query language implemented in C++ for Windows, Mac OS X and Linux. Negation is allowed in heads and bodies of rules as well as in recursion	GNU GPL v3
	RDFox ^[21]			RDF triple store with Datalog reasoning. Implements the FBF algorithm for incremental evaluation.	custom licence, free for non-commercial use ^[22]
	Souffle ^[23]			an open-source Datalog-to-C++ compiler converting Datalog into high-performance, parallel C++ code, specifically designed for complex Datalog queries over large data sets as e.g. encountered in the context of static program analysis	UPL v1.0
	Cascalog [🔗]		Hadoop	a Clojure library for querying data stored on Hadoop clusters	Apache
Clojure	Clojure Datalog [🔗]			a contributed library implementing aspects of Datalog	Eclipse Public License 1.0
	Crux [🔗]	Yes [🔗]	Apache Kafka	A general-purpose database with an "unbundled" architecture, using log-centric streaming of documents and transactions to achieve significant architectural flexibility and elegant horizontal scaling. Pluggable components include Kafka, RocksDB and LMDB. Indexes are bidirectional to support point-in-time Datalog queries by default. Java and HTTP APIs are provided.	MIT License
	Datscript [🔗]		in-memory	Immutable database and Datalog query engine that runs in the browser	Eclipse Public License 1.0
	Datavein [🔗]		LMDB	A fork of Datscript that is optimized for the LMDB durable storage	Eclipse Public License 1.0
	Datatrike [🔗]		file, in-memory	A fork of Datscript with a durable backend using a hitchhiker tree [🔗] .	Eclipse Public License 1.0
Erlang	Datalog [🔗]			The library is designed to query and formalise relation of n-ary streams using datalog. It implements an ad-hoc query engine using simplified version of general <i>logic programming</i> paradigm. The library facilitates development of data integration, information exchange and semantic web applications.	Apache v2
Haskell	Dyna ^[24]			Dyna is a declarative programming language for statistical AI programming. The language is based on Datalog, supports both forward and backward chaining, and incremental evaluation.	GNU AGPL v3
	DDiog ^[25]			DDiog is an incremental, in-memory, typed Datalog engine. It is well suited for writing programs that incrementally update their output in response to input changes. The DDiog programmer specifies the desired input-output mapping in a declarative manner, using a dialect of Datalog. The DDiog compiler then synthesizes an efficient incremental implementation. DDiog is based on the differential dataflow ^[26] library.	MIT License
Java	AbcDatalog ^[27]			AbcDatalog is an open-source implementation of the logic programming language Datalog written in Java. It provides ready-to-use implementations of common Datalog evaluation algorithms, as well as some experimental multi-threaded evaluation engines. It supports language features beyond core Datalog such as explicit (dis-)unification of terms and stratified negation. Additionally, AbcDatalog is designed to be easily extensible with new evaluation engines and new language features.	BSD
	IRIS ^[28]			IRIS extends Datalog with function symbols, built-in predicates, locally stratified or un-stratified logic programs (using the well-founded semantics), unsafe rules and XML schema data types	GNU LGPL v2.1
	Jena			a Semantic Web framework which includes a Datalog implementation as part of its general purpose rule engine, which provides OWL and RDFS support ^[29]	Apache v2
	Socialite ^[30]			Socialite is a datalog variant for large-scale graph analysis developed in Stanford	Apache v2
	Graal ^[31]			Graal is a Java toolkit dedicated to querying knowledge bases within the framework of existential rules, aka Datalog+/-.	CeCILL v2.1
	Flix [🔗]	Yes [🔗]		A functional and logic programming language inspired by Datalog extended with user-defined lattices and monotone filter/transfer functions.	Apache v2
Lua	Datalog ^[32]	Yes ^[33]		a lightweight deductive database system.	GNU LGPL
OCaml	datalog [🔗] ^[34]			An in-memory datalog implementation for OCaml featuring bottom-up and top-down algorithms.	BSD 2-clause
Prolog	DES [🔗] ^[35]			an open-source implementation to be used for teaching Datalog in courses	GNU LGPL
Python	pyDatalog [🔗]		11 dialects of SQL	adds logic programming to Python's toolbox. It can run logic queries on databases or Python objects, and use logic clauses to define the behavior of Python classes.	GNU LGPL
Racket	Datalog for Racket ^[36]				GNU LGPL
	Datafun ^[37]			Generalized Datalog on Semilattices	GNU LGPL
Ruby	bloom [🔗] / bud			A Ruby DSL for programming with data-centric constructs, based on the Dedalus [🔗] extension of Datalog which adds a temporal dimension to the logic.	BSD 3-Clause
Rust	Crepe [🔗]			Crepe is a library that allows you to write declarative logic programs in Rust, with a Datalog-like syntax. It provides a procedural macro that generates efficient, safe code and interoperates seamlessly with Rust programs. It also supports extensions like stratified negation, semi-naïve evaluation, and calling external functions within Datalog rules.	MIT License / Apache 2.0
	Datafrog [🔗]			Datafrog is a lightweight Datalog engine intended to be embedded in other Rust programs.	MIT License / Apache 2.0
Tcl	tcldbdd [🔗] ^[38]			Implementation based on binary decision diagrams . Built to support development of an optimizing compiler for Tcl.	BSD
Other or Unknown Languages	bddbldb ^[39]			an implementation of Datalog done at Stanford University . It is mainly used to query Java bytecode including points-to analysis on large Java programs	GNU LGPL
	ConceptBase ^[40]			a deductive and object-oriented database system based on a Datalog query evaluator : Prolog for triggered procedures and rewrites, axiomatized Datalog called « Telos » for (meta)modeling. It is mainly used for conceptual modeling and metamodeling	BSD 2-Clause

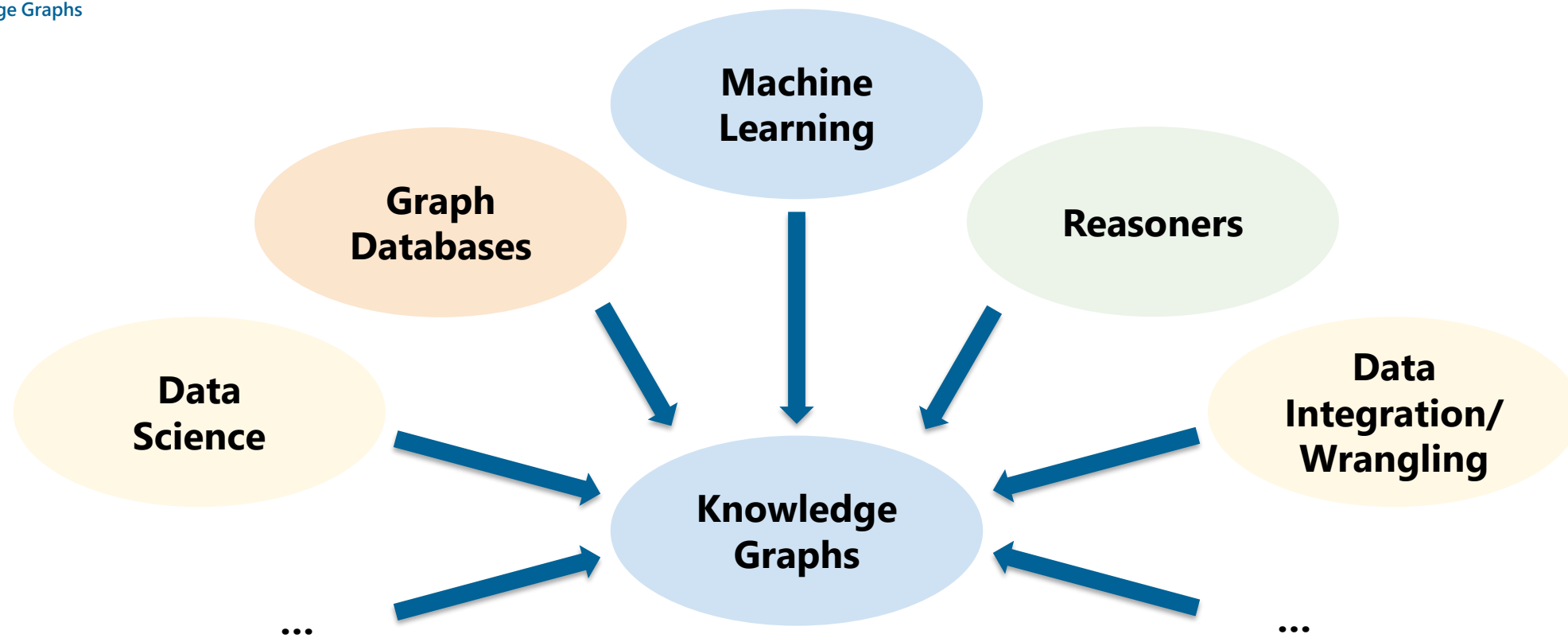




Datalog with Creation

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Racket	Datalog for Racket^[36]			





Azure Cosmos DB



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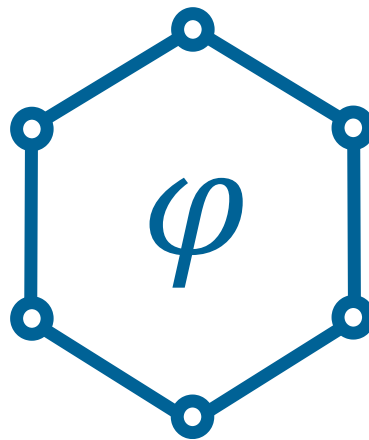
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